

Machinoia, Machine of Multiple Me: Integrating with Past, Future and Alternative Selves

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Figure 1: Machinoia, a wearable installation that extends the user with two additional heads each of which are unique variations of the users identity: who you once were, and who you'll eventually become.

ABSTRACT

Our body and mind relate in ways which are extraordinarily enigmatic and seemingly incomprehensible. Recent findings exemplify this by showing not just that our minds can phenomenologically inhabit multiple bodies but also that our body can be accessed by multiple minds. As an exploration of this concept, we present Machinoia, a symbiotic augmentation that extends the user with two additional heads each of which are unique variations of the users identity: who you once were, and who you'll eventually become. We used a generative adversarial network to synthesize life-like human faces and controlled them through artificial attitude models extracted from social media data of the wearer, thus creating “artificial personal intelligences” of the wearer, bringing to life past and future versions of oneself.

CCS CONCEPTS

• **Computer systems organization** → **Embedded and cyber-physical systems**; *Real-time systems*; • **Human-centered computing** → **Interaction paradigms**; **Human computer interaction (HCI)**.

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KEYWORDS

human-computer integration, wearable technology, artificial intelligence, philosophy of mind, human enhancement

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1 INTRODUCTION

Can one body be inhabited by multiple minds? The relationship between our body and mind is an old yet still incredibly puzzling phenomenon. It seems that our experiences of the world are intimately connected to our body, in the sense that physiological states influence our mental state and vice versa. This supposed interdependence of phenomenological and physiological states suggests a strong bond between our body and the mind. However, recent literature challenges this by showing that the mind-body relationship can be flexible in various ways [1, 2, 23, 25]. As an example, Craig Murray et. al. [23] were able to demonstrate that participants in virtual reality not only could successfully embody human-like avatars but also non-anthropomorphic ones such as that of a lobster. Further, one study also showed that not only can one's mind inhabit a different body but even multiple bodies simultaneously. In the study, participants were able to control and perceive up to four bodies at the same time while reporting a Sense of Body-Ownership and Agency for each body [20]. These results



Figure 2: Wearing Machinoia: the wearer can embody and communicate with future and past selves.

suggest that our body and mind are not as strictly tied together as one might have originally thought.

This entanglement is further perplexed when considering a multiplicity of minds and bodies. An example of this is that of conjoined twins, who, despite being thought of as two individuals, often are found to have overlapping sensations and movements [22]. The conjoined twins Krista and Tatiana Hogan, for instance, share senses of touch and taste and even control one another’s limbs. Tatiana can see out of both of Krista’s eyes, while Krista can see out of one of Tatiana’s; Krista is able to experience distaste towards food that Tatiana is eating and will try to prevent her from eating it; and the twins even report that they talk to each other in their heads [4, 8, 26]. The Hogan twins acquired these capabilities due to a thalamic bridge connecting them in their brains [5]. Together with the prior examples, this demonstrates that not only can one’s mind adjust and inhabit multiple bodies, but one’s body can also be shared between multiple minds. The phenomenon of having more than one head is referred to as polycephaly.

1.1 Polycephaly

Polycephalic occurrences, though, are not just extended to cases of conjoined twins, there are multi-headed entities appearing both in mythology and pop-culture throughout human history. Famous examples include the Hydra, a serpentine water monster from Greek mythology [6] and its modern reinterpretation, Ghidorah in the Godzilla film franchise; or Cerberus, a multi-headed canine monster that guards the gates of the underworld. Further, often multiple heads personify different characteristics, personality traits or functions, and having these multiple minds is often perceived as a trait of increased wisdom. This is exemplified by Hindu deities such as Agni, the God of fire, Brahma, the creator God, and Kartikeya, the God of war, all often depicted having multiple heads [11, 32]. As another example, the ten heads of the Hindu demon-king Ravan each contain specific branches of knowledge said to make Ravan one of the most learned scholars and strategist to have ever lived [24]. Instead of trying to fit all knowledge into one mind, he has it distributed across multiple, utilizing these extra minds for some sort of externalization of cognitive abilities. Artist Krzysztof Wodiczko also explores this kind of polycephalic system in his piece “Aegis” [14, 28], where the wearers engages in a dialogue about their homeland and identity.

1.2 Cognitive Externalization

This kind of cognitive externalization is not just something mythical but actually a scientifically studied phenomenon also known as cognitive offloading [27], meaning cognitive tasks are externalized to reduce mental load. This is actually already an integral part of how we think and function. A recent study on memorization showed that using the internet to answer questions primed retentive use, and more accurate and faster answers. In the study, participants who used search engines to answer sets of trivia questions were more quicker, accurate, and likely to rely on search engines for question answering than participants who answered the questions from memory [31]. Similarly, another study involving an AI system capable of analyzing the quality of arguments found that participants’ reasoning outcomes were improved by helping users differentiate between statements supported by evidence and without [7]. This suggests that using technological means external to the mind for cognitive tasks such as memory or reasoning can enhance reaction time and quality, inspiring more intimately integrated cognitive structures between an artificial and human mind.

The notion of task-distribution between multiple humans and computer agents is further supported in Thomas W. Malone’s concept of ‘Superminds’. According to Malone, the networks of humans-humans and humans-computers collectively sharing information and tasks to arrive at co-decisions can be compared to how the brain distributes tasks between specialized brain regions for effectively completing goals. He argues that such networks lead to increased intelligence of individuals and collectives; what he calls ‘Superminds’ [18].

According to Yuval Noah Harari, this is just the beginning. In his book “Homo Deus”, he regards utilizing the internet for cognitive externalisation as a first step towards a future of human-machine symbiosis where technology will not only assist us externally but eventually merge with us and make us “Homo Deus”, or God

Humans, endowed with a mix of artificial and human abilities such as the immense accumulated knowledge of all humankind [9].

1.3 Introspection and Authenticity

We believe that an intimate human-machine self¹ will not only allow us to embody the vast external knowledge of the digital space, but it will just as much help us discover knowledge internal to ourselves, giving us insights on our personality, helping us reflect upon ourselves and our past experiences, and enhancing our capabilities for introspection.

Doing this, we believe, will make us better at acknowledging ourselves and making decisions about our future. In “Being and Time”, the philosopher Martin Heidegger expresses the possibility that changes in identity over time are a consequence of external influences [10]. Changing between external contexts and assuming new roles in society alters our identity, but it often does so, in Heidegger’s terms, “inauthentically” [13] [10]. This is because we often don’t use our past experiences when we make decisions in a new and unfamiliar environment; we often just adapt to the environment and go along with things; we try to blend in as easily as possible. However, in order to live authentically to ourselves, Heidegger claims, we need to conform our past experiences with our present; we need to be authentic to our personal historicity by letting it influence and bare significance in everything we do. We believe, that polycephalic systems can play a significant role in bringing about this authenticity by letting our past and potential future selves saturate ourselves in the present. Being able to consult our past or future selves enables us to remember our past, get perspectives on our future, and possibly live a more authentic life.

2 DESIGN

Inspired by the ideas of human-machine symbiosis, polycephaly, and the notion of conforming our past and future experiences with our present environment, we explored the idea of a single self being distributed across multiple minds and temporal identity-variations (past, present and future).

As a manifestation of this, we present Machinoia, a symbiotic augmentation that extends the user with two additional heads each of which are unique variations of the users identity: who you once were, and who you’ll eventually become. The augmentation consists of two interactive screens with two necks connecting to the shoulders of the wearer. We use screens to act as a metaphorical mirror that show the multiple faces of the user reflecting the future, past, or potentially alternative versions of oneself. The life-like generated faces of the wearer can communicate and express different facial expressions based on their interaction with the wearer, turning the wearer into a digitally augmented polycephalic being.

3 TECHNICAL IMPLEMENTATION

To realistically generate human faces of the wearer, representing multiple versions of the user (younger and older), we deployed a generative adversarial network (GAN), more specifically “StyleGAN”.

¹We are aware of the vast literature and complexity of the term ‘Self’ (like Identity, Sense of Self, and so on). However, due to the speculative nature of this paper, we use the term ‘Self’ loosely to indicate the many potential implications our system might have on a person’s ‘Selfhood’

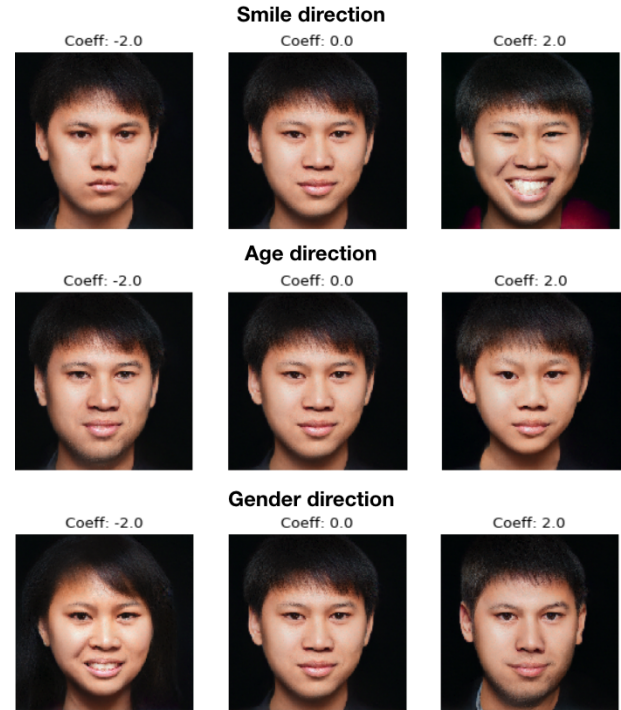


Figure 3: Alternative versions of the wearer generated by StyleGAN with feature specific vectors applied to them.

The StyleGAN model used in the project consists of a generator and discriminator architecture that was pretrained on a large dataset of human faces² [12], thus allowing the model to learn the features and create a universal latent representation of human faces. The model was given a picture of the wearer to reverse-generate a vector representation. This vector representation could be influenced to generate alternative versions of the reverse-generated face for different ages, genders, and other features by applying feature specific vectors. In our implementation, we used the StyleGAN model to generate older and younger versions of the wearer by applying an age vector to the model. The generated images of each face were then animated through the state-of-the-art First Order Motion Model generator network [30], which utilizes a self-supervised learning algorithm to determine the local transformation of facial animation to generate video outputs with realistic and complex motion. To generate the video output, we pre-recorded template videos for different emotional responses which then drives the StyleGAN generated images of older and younger versions of the wearer’s face. The resulting facial animation is then used as the visual response to the wearer, giving them the sensation of talking to a real human face.

Creating these alternative versions of the wearer from his/her past and future, we were inspired by the work of Liu and Maes [17] that demonstrated how to employ a computational approach for extracting a person’s attitude from their personal corpus of textual information. Based on their approach, we developed a method to

²We use the following StyleGAN encoder from <https://github.com/Puzer/stylegan-encoder>

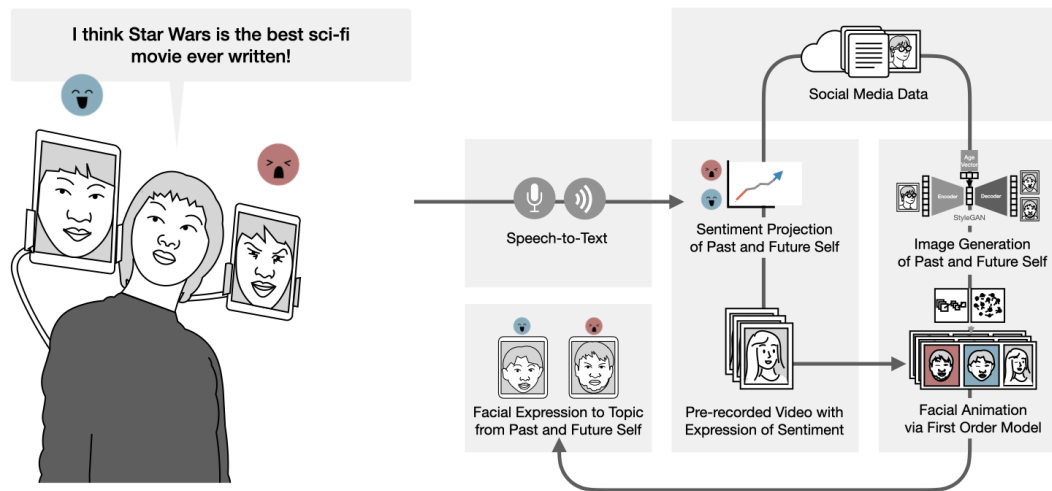


Figure 4: Left: A full size view of the Machinoia system. Right: A diagram of the system. The process begins when the wearer talks to the heads. The speech is then recognized and processed into text, which uses semantic word embedding to extrapolate the attitudinal response of the head, which lastly is expressed through facial animation.

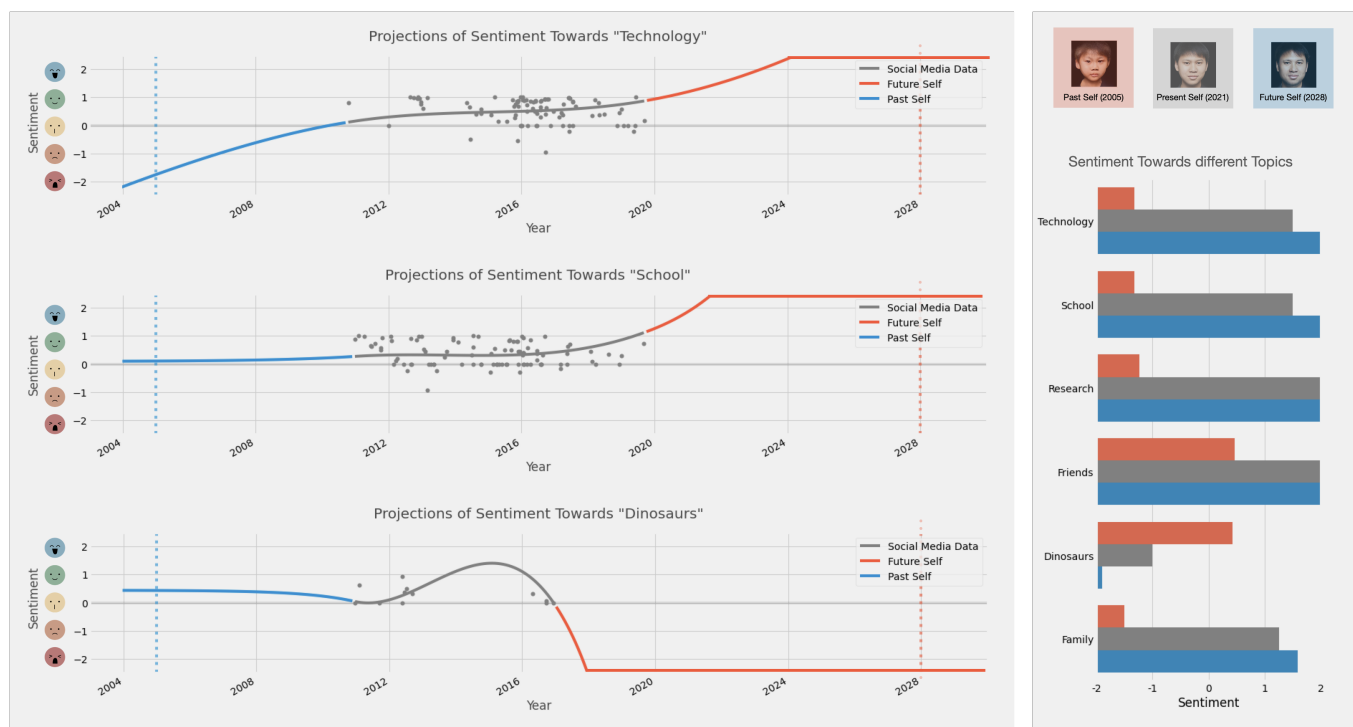


Figure 5: Projections from a Seasonal AutoRegressive Integrated Moving Average with exogenous variables (SARIMAX) model trained on a 3. order polynomial fit over the wearers social media data. Left: Past and Future projections of sentiment towards "Technology", "School", and "Dinosaurs". The vertical dotted lines show the Past (blue) and Future (red) Self's attitude. Right: Bar chart with the attitudes of Past, Present and Future Self towards different topics

construct a computational model of the wearer’s attitude on different topics using longitudinal social media information obtained with the consent of the wearer. In our implementation, we extracted 10 years of one of the authors of this paper’s Facebook social media data (2010 to 2020). We preprocessed the data by filtering stop words and non-ASCII encoded characters. We extracted the key ideas inside of each social media post using the NLTK natural language processing library [3] and by measuring the sentiment of the person toward a specific topic over time.

For example, in a post “I think Star Wars is the best sci-fi movie ever written in the original trilogy with tons of original moments, and familiar ones packed into one package”, we can extract keywords such as “sci-fi”, “original trilogy”, “original moments”. Using name identity recognition to identify the topic of conversation, we can also extract sentiment values (pos: 0.7/neg: 0.3), which together with the identified topic could demonstrate whether the wearer may like a certain topic or not, in this case science fiction movies like Star Wars.

We measure the changes of the sentiment on different topics at different ages of the wearer to estimate values for an older version of the wearer. The trajectory of the sentiment changes are then fitted to a polynomial graph, which allows us to extrapolate and speculate about the person’s attitudes in the future or past. To extrapolate these attitudes we implemented a Seasonal AutoRegressive Integrated Moving Average model with exogenous variables (SARIMAX) trained on the polynomial fit over the wearers social media data. The resulting “speculative attitude model” is then used to trigger different responses upon usage when the wearer talks to his/her “artificial personal intelligent” personas.

The speculative attitude model containing these sentiment values is then embedded in each digital head using the iOS swift developer environment. Utilizing this, each face is an iOS app on a separate device, with each including a speech to text module, a semantic word embedding model, and a speculative attitude model. When a face hears the voice of the wearer, the speech to text translates the wearer’s question or query into text, which gets computed into a set of semantic vectors. These vectors are then used to predict a personal response by calculating cosine similarity between the topic in the query and the topic in the speculative model. If the response is positive, then the digital face will smile, and if it is negative it frowns, and so on. The degree of expression is determined by the sentiment values contained within the speculative attitude model.

4 POTENTIAL IMPACT

Thinking beyond the realisation of the Machinoia system, we see opportunities and limitations of such symbiotic systems in the future.

4.1 Opportunities

In the future, we envision Machinoia to enable a greater awareness and conformity of past, present and future experiences with our beliefs and decision making. With past and future versions always being available and responsive for consultation, we believe that Machinoia will a greater degree of perspective-taking and introspection as it, in Heideggerian terms, allows us to be more authentic to ourselves by being reminded of our personal history, goals, and

potential future projections. As eidetic mirrors, we thus consider Machinoia to enable a kind of reflection from talking to yourself and understanding of consequences of your actions leading to a more realistic perception of, for instance, risks and thus to positive behavioral change.

The idea of talking to different versions of yourself is further explored in a recent article by Şenel & Slater [29] where a dialogue between the present self and the future self was facilitated through virtual reality. In a proposed study, they expect that a present smoking self consulting a future smoking cessation self and future still smoking self will lower nicotine dependence and lead to long-term healthy behaviour choices. Similar to Şenel & Slater’s work, we expect increased positive healthy development and behavioral change as results from using Machinoia as a guide to promote healthy adolescent development in society.

Further, while we showed the extended heads to be young and old versions of the wearer, there are many other interesting applications of this technology. For example, switching one’s gender, political leanings, or ethnicity can help people develop a deeper understanding of humanity and the world by having multiple perspectives and attitudes inhabit the same body. Additionally, the different heads might not be restricted to the attitude of one persona. Attitudes about issues could be mined from different groups online, enabling a person to see the world from a “Hive Mind” perspective and contrast it with their own lived experience and resulting beliefs and attitudes.

4.2 Limitations

In addition to presented opportunities, we also see additional concerns arising from using Machinoia in the future. For instance:

- Is it safe to “encourage” people to live in the past/future rather than enjoy the present with their actual wisdom?
- What happens when a person is faced with trauma from a past self?
- How much can a person affect and change the future self? Will personal progress be halted if not?

People change over time and often that change is good. For example, someone may overcome some challenges or ill-informed opinions on something by learning and “growing”, and would not wish to recollect such situations anymore. There is a risk that by offering the younger version’s perspective, the person would not move on and get over their past issues.

Also, there is a risk that the person could be “overwhelmed” with perspectives from the different heads and not be able to make decisions, e.g. if the different heads disagree on a decision or issue.

Further, as the future self is a projection made from past events, negative or less beneficial experiences could further significantly bias the projected future self adversely despite recent (but small) positive personal progress.

By shuttling between the past and the future, there may be a danger of not being in the present while making certain decisions. Over a period of time, this can limit the growth potential of a person and the past and future may not be very different.



Figure 6: Machinoia demonstrates a new type of venture into the concept of intimate human-machine symbiosis, namely the approach of a multiple mind integrated polycephalic system.

5 CHALLENGING THE HUMAN-MACHINE DICHOTOMY

Recent developments within HCI are starting to challenge the implicit notion of humans and computers as two separate entities and investigate ways in which humans and computers can integrate “as one human-technology assemblage” [21]. We consider Machinoia an example of such integration, and aim to promote the idea of designing devices with cognitive distribution across multiple minds and temporal modalities (past, present and future), in this work exemplified as multiple heads. Learning from the fields of the cognitive sciences and extraordinary examples like conjoined twins, we believe technology can play a more direct and intimate role in the relationship between our body and mind.

Our work is in many ways a reinterpretation of J.C.R. Licklider’s vision of human-machine symbiosis[16]. While to date humans and computers are still regarded as separate entities, technology is moving ever closer to us and gradually becoming indispensable (e.g. cognitive offloading for critical tasks). Certain tracks in both HCI and AI focus on the vision of an integrated human-machine self with cognitive distribution across multiple minds [19, 21, 33]. Going beyond Licklider’s original vision, we are shifting from “interaction” between two entities to a more intimate “integration”, where both human minds and artificial minds seamlessly integrate and constitute “one human-technology assemblage” [15, 21].

We are increasingly becoming dependent on our technological counterparts to think and act. In a way, Machinoia exemplifies both a critique and a praise of this human dependency on technology, which seems to have both positive and negative consequences, questioning whether in the future we will see ourselves as a species that is reliant on machines or as a new breed, “Homo Deus”, endowed with a mix of artificial and human capabilities. The purpose of Machinoia is then to question the relationship between human and machine identities, and whether the self is a singular thing or is multiplicitous.

6 CONCLUSION

Machinoia demonstrates a new type of venture into the concept of human-machine symbiosis, namely integrated polycephalic multiple mind systems. Based on the extraordinarily ambiguous connection between our mind, body and the external world, Machinoia

utilizes the extension of two heads to externalize and recall past and potential future experiences and attitudes. Through this temporal merging, Machinoia helps us discover knowledge internal to ourselves across multiple temporal modalities, giving us insights into our different personality traits, and enhancing our overall capability for introspection, so that we may live a life that according to Heidegger is more authentic to ourselves. Machinoia aims to raise awareness of the gradual adoption of technology as an integrated part of ourselves, encouraging discussion of its positive and negative potential impact on what it means to be Human(s).

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